

SANTA'S SLEIGH

A small sleigh. A biological engine.

A narrative engineering reconstruction of the Christmas Eve vehicle system.



STEP ONE

THE SERVICE, NOT THE MYTH

Functional unit, reference flow and modeled cut-off are frozen before any calculation.



FUNCTIONAL UNIT

One reusable airborne gift-delivery sleigh completing one Christmas Eve mission.

REFERENCE FLOW

1/12 sleigh system + one-night reindeer motive system + staging support.

BOUNDARY

Materials, repair, forage and local support. Gifts and chimneys are excluded.

STATUS

Narrative reconstruction: documented sources, engineering assumptions and proxies.



The model does not measure Christmas. It measures the physical system required to stage the legend.

STEP ONE

BOUNDARY BEFORE WONDER

The sleigh is treated as a reusable product system plus a biological motive system.

INCLUDED

Sleigh materials
Harness set
Annual repair
Reindeer forage
Workshop-to-staging van

EXCLUDED

Toy manufacturing
Household chimney entry
Workshop construction
Magic flight energy

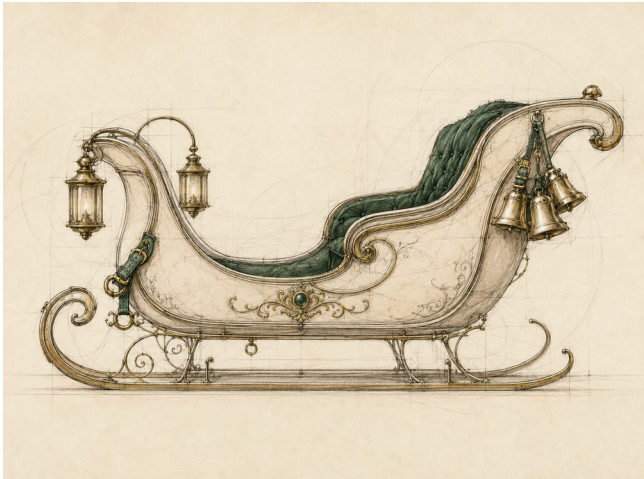


Cut-off event: the model stops when the mission-ready sleigh leaves staging with gifts already outside the boundary.

STEP TWO

RECONSTRUCT THE VEHICLE

A single canonical design is carried through every internal page.



1.55 m
height

4.2 m modeled length

BODY 160 kg laminated wood	RUNNERS 40 kg steel proxy	HARNESS 30 kg leather/textile	LIFETIME 12 Christmas missions
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The sleigh is modeled as durable infrastructure: the vehicle is amortized, the motive system is counted per mission.

STEP TWO

INVENTORY IN MASS

Every line has a quantity, a basis and a factor route.



Flow	Qty	Unit	Basis
Wood body + rails	160	kg	engineered
Steel runners + brackets	40	kg	engineered
Harness set	30	kg	proxy
Bells + fittings	8	kg	proxy
Finishes + small parts	5	kg	proxy
Repair materials	6	kg	annual
Reindeer forage	48	kg	literature
Staging van support	30	km	proxy



Gift manufacturing is not hidden in the sleigh result; it is excluded rather than guessed.

STEP TWO

FROM WORKSHOP TO MISSION

The manufacturing model stays inside the declared boundary.

**01 SHAPE THE WOODEN FRAME**

Laminated ash/oak-like members carry the mass model.

02 FORGE RUNNERS AND FITTINGS

Steel and mixed metal proxies cover runners, brackets and bells.

03 STITCH HARNESS AND UPHOLSTERY

Leather/textile is modeled with DEFRA clothing proxy.

04 PREPARE THE TEAM

Forage converts biological motive energy into a counted flow.



No miracle is assigned an emission factor; only physical materials and support flows enter the total.

STEP TWO

ENERGY BECOMES FORAGE

The flight has no modeled fuel, but the reindeer team has a material input.



TEAM

8 reindeer
canonical source

FEED

6 kg/head/day
baseline

FORAGE

48 kg
one mission

STAGING

30 km van
local support



The modeled hotspot is not airborne propulsion; it is the one-night biological supply flow.

STEP THREE

FACTORS AND PROXIES

DEFRA 2026 is used first; external sources support biology and sensitivity context.

ID	Flow	Unit	Factor
S1	Wood, primary	kg CO2e/kg	0.270
S2	Steel/metal proxy	kg CO2e/kg	2.862
S3	Harness clothing proxy	kg CO2e/kg	22.310
S5	Average plastics	kg CO2e/kg	3.171
S6	Food and drink proxy	kg CO2e/kg	3.701
S7	Van freight support	kg CO2e/km	0.280

PROXY DISCLOSURE

Food and drink is used as a conservative proxy for forage. Clothing is used for harness because DEFRA 2026 has no dedicated leather-harness factor. Source IDs link to the workbook register.

➔ The source register is outside the artwork; short labels here are only reading aids.

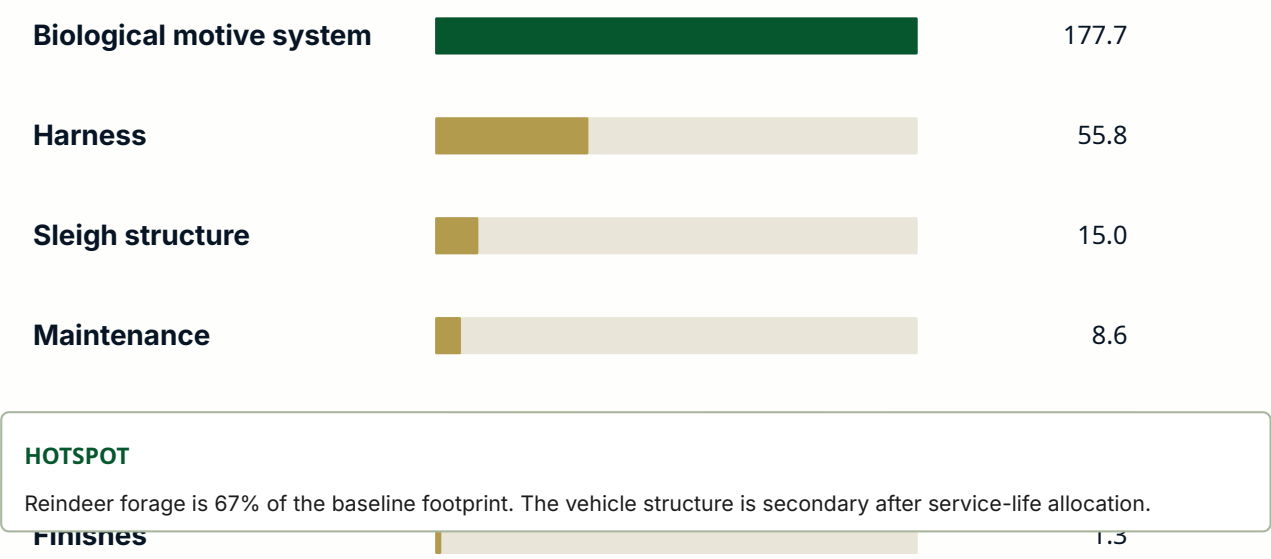
STEP THREE

THE BASELINE RESULT

Total first. Contributions second. Interpretation third.

267

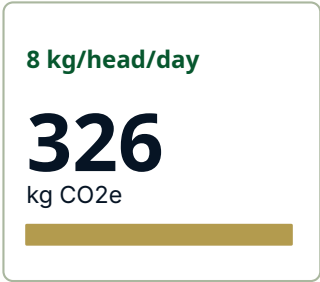
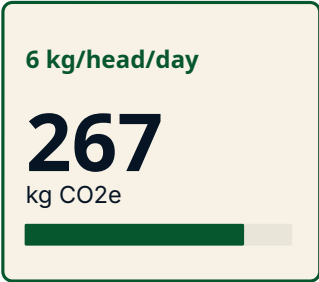
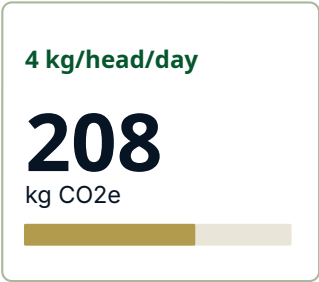
kg CO2e per Christmas Eve mission



STEP FOUR

SENSITIVITY: FEEDING RATE

One physical assumption changes: kilograms of forage per reindeer per mission day.



Within the documented 4-8 kg/day range, the result moves from 208 to 326 kg CO2e.

STEP FOUR

SENSITIVITY: HARNESS FACTOR

One data assumption changes: the emission factor assigned to the leather/textile harness.



Re-used harness proxy

211

kg CO2e

DEFRA re-used clothing

Baseline clothing proxy

267

kg CO2e

DEFRA primary clothing

Leather literature context

351

kg CO2e

External leather proxy,
Carbonfact cow bovine average



The leather/harness proxy is material: switching from reused proxy to leather-specific context shifts the total by more than 140 kg CO2e.

FINDING

WHAT THE MODEL TELLS US

Four claims, all derived from the same baseline and sensitivity evidence.

**01 The sleigh is not the main burden.**

After lifetime allocation, structure and finishes remain below the forage hotspot.

02 Biology replaces fuel.

No fuel is modeled for flight, but feed becomes the motive input.

03 Proxies are decisive.

Harness results change materially when the leather/textile factor changes.

04 The boundary protects the answer.

Gifts are excluded; otherwise the episode would measure Christmas retail, not the sleigh.

The miracle is boundary-free; the model is not.

MYTHS & LEGENDS

The impossible becomes useful when assumptions
become visible.

SANTA'S SLEIGH